

Octonions, G_2 , and Knot Topology in the Arithmetic of Order Framework: A Topological–Algebraic Reinterpretation of the Higgs Mechanism within the Standard Model

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Dedication

This work is dedicated to the foundational pioneers of octonionic and exceptional algebraic approaches to physics, whose groundbreaking insights have paved the way for modern formulations connecting algebra, topology, and fundamental interactions.

As detailed in the Appendix, the profound contributions of researchers such as Catto, Chesley, Günaydin, Piron, and Ruegg continue to inspire and guide the development of the Arithmetic of Order framework.

Abstract

We present a unified framework connecting knot theory, exceptional algebra, and particle physics that provides a *topological–algebraic reinterpretation* of the Higgs mechanism within the Standard Model (SM). The SM gauge groups $U(1)$, $SU(2)$, and $SU(3)$ arise naturally from the three Reidemeister moves on knots and braids, with a rigorous Lie-algebraic identification via Schiller’s strand–tangle model. These gauge symmetries embed into the exceptional Lie group G_2 , the automorphism group of the octonions, and are realized geometrically through coset spaces such as $G_2/(SU(2)_L \times SU(2)_R)$.

Within the Arithmetic of Order (AoO) framework, the Higgs field is reinterpreted not as a fundamental scalar particle but as an *internal algebraic phase transition* on this coset manifold. This phase transition triggers symmetry breaking, producing W and Z boson excitations consistent with collider data. The framework yields a fully computable, discrete physical model that recasts the SM as an inevitable consequence of topology, exceptional algebra, and discrete geometry.

1 Introduction

The Standard Model (SM) successfully describes electromagnetic, weak, and strong interactions via the gauge groups $U(1)$, $SU(2)$, and $SU(3)$. Yet the *origin* of these symmetries and the nature of the Higgs mechanism remain foundational open questions.

This work develops a conceptually novel, mathematically rigorous framework that:

- Roots SM gauge symmetries in knot theory, identifying the three Reidemeister moves as generators of $U(1)$, $SU(2)$, and $SU(3)$,
- Embeds these symmetries within the exceptional Lie group G_2 , the automorphism group of the octonions,
- Reinterprets the Higgs field as an *internal algebraic phase transition* on the coset space $G_2/(SU(2)_L \times SU(2)_R)$,
- Grounds all constructions in the Arithmetic of Order (AoO) framework, which models the universe as a finite, computable structure.

This topological–algebraic viewpoint offers a striking alternative to conventional Higgs physics, eliminating the need for a fundamental scalar and naturally aligning with discrete computational physics.

Finite yet unbounded scalar base: In this framework, the octonion algebra is defined over a finite, yet *unbounded*, set of rational scalars $\mathbb{Q}$. Unlike the standard construction over $\mathbb{R}$ or $\mathbb{C}$, this discrete scalar base reflects the Arithmetic of Order (AoO) philosophy, where $\mathbb{Q}$ is a countable, computable field sufficient to encode physical information without invoking the uncountable continuum. This approach reconciles finiteness with arbitrary precision, aligning with AoO’s foundational premise that the universe is fundamentally a finite, computable structure.

Contributions. This work establishes a complete representation–theoretic realisation of the Standard Model gauge group $U(1) \times SU(2) \times SU(3)$ within the finite, computable Arithmetic of Order (AoO) framework. The three gauge factors emerge as the minimal topological closure of AoO’s discrete arithmetic via the Farey– $SL(2, \mathbb{Z})$ recursion lifted to the braid group B_3 , with each Reidemeister move mapped to explicit 2×2 or 3×3 matrix representations whose commutators satisfy the Jacobi identities. These symmetries are embedded into the exceptional group $G_2 = \text{Aut}(\mathbb{O}_{\mathbb{Q}})$, where the Higgs sector is reinterpreted as an internal algebraic phase transition on the coset

$$G_2/(SU(2)_L \times SU(2)_R) \longrightarrow \mathcal{H}_{\text{AoO}} \subset \mathbb{O}_{\mathbb{Q}},$$

a finite quaternionic–octonionic subspace that reproduces the familiar $\mathbb{C}^2$ doublet only in the low–energy continuum limit. The model incorporates Moufang torsor structures as non–associative corrections to the Higgs potential and yields concrete, falsifiable predictions for Higgs–gauge couplings, electroweak observables, and rare decay rates — all without invoking a fundamental scalar field or continuum spacetime at the microscopic level.

2 Reidemeister Moves and Gauge Symmetries

Each Reidemeister move corresponds precisely to an SM gauge group:

Move	Topological Action	Gauge Group
R1 (Twist)	Single-strand rotation	$U(1)$
R2 (Poke)	Two-strand crossing	$SU(2)$
R3 (Slide)	Three-strand sliding	$SU(3)$

Schiller’s (2025) strand–tangle model rigorously constructs explicit algebraic generators for these moves, demonstrating their closure under the corresponding Lie algebras. This places the topology–gauge correspondence on firm algebraic footing.

3 Octonions and G_2 Embedding

The octonions $\mathbb{O}$ form an eight-dimensional normed division algebra whose automorphism group is the exceptional Lie group G_2 . Crucially,

- $SU(3) \subset G_2$ acts as the color symmetry,
- $SO(4) \cong (SU(2)_L \times SU(2)_R)/\mathbb{Z}_2$ encodes the electroweak symmetry.

Schiller’s algebraic generators embed explicitly into these subalgebras, unifying knot topology and SM gauge symmetries within G_2 .

4 Geometric Realization and the Higgs Field

4.1 Coset Space Structure

Within the Arithmetic of Order (AoO) framework, we reinterpret the Higgs field not as a fundamental complex scalar field, but as a coordinate $\phi(x)$ on a discrete octonionic coset manifold:

$$G_2/(SU(2)_L \times SU(2)_R) \longrightarrow \mathcal{H}_{\text{AoO}} \subset \mathbb{O}_{\mathbb{Q}},$$

where $\mathcal{H}_{\text{AoO}}$ is a finite-dimensional quaternionic–octonionic subspace constructed from AoO’s finite computable algebra.

The familiar continuum description of the Higgs doublet as an element of $\mathbb{C}^2$ emerges only as a low-energy, effective approximation:

$$\mathcal{H}_{\text{AoO}} \xrightarrow{\text{continuum limit}} \mathbb{C}^2,$$

thus preserving AoO’s core principle of no fundamental $\mathbb{R}$ or $\mathbb{C}$.

4.2 Effective Lagrangian and Couplings

In terms of strand variables, the effective Higgs-sector Lagrangian is

$$\mathcal{L}_\phi = \frac{1}{2}(\partial_\mu \phi)^2 - V(\phi) + \bar{\psi}\Gamma(\phi)\psi + \dots,$$

where $V(\phi)$ encodes topological rearrangement energies and $\Gamma(\phi)$ governs couplings to the W and Z bosons.

4.3 AoO Interpretation

- The Higgs field is not fundamental but arises as an emergent algebraic phase transition within octonionic coset geometry.
- The observed 125 GeV Higgs “signal” corresponds to gauge boson excitations emerging from this internal transition.
- This approach fits naturally within a fully discrete, computable physical universe described by the AoO framework.

5 Torsors, Moufang Loops, and Octonionic Geometry

Recent developments in projective geometry by Bertram and Kinyon [6] provide crucial mathematical foundations for the non-associative algebraic structures central to our AoO framework. Their work interprets torsors and ternary Moufang loops in classical projective geometry, elucidating the algebraic and geometric underpinnings of octonionic multiplication and the exceptional group G_2 .

- The automorphism group G_2 preserves the octonion multiplication satisfying the Moufang identities:

$$(xy)(zx) = x(yz)x,$$

which identify Moufang loops as natural algebraic structures associated with non-associative geometry. Bertram and Kinyon show how such Moufang structures arise naturally in non-Desarguesian projective planes [6, Theorem 5.4], consistent with the coset $G_2/(SU(2)_L \times SU(2)_R) \cong S^6$ that models the Higgs sector in our framework.

- Their construction of torsors (principal homogeneous spaces) models symmetry breaking patterns as discrete algebraic phase transitions on cosets, perfectly matching the AoO interpretation of the Higgs as a torsor morphism rather than a fundamental scalar field [6, Section 3].
- For the Desarguesian (associative) case, classical gauge groups emerge naturally, whereas the Moufang (octonionic, non-associative) case introduces new geometric features consistent with corrections around the Higgs scale (125 GeV) [6, Section 4].

- The Higgs potential $V(\phi)$ can be enriched by Bertram–Kinyon’s ternary operations $T(x, y, z) := (xy)z$, leading to terms in the Lagrangian of the form:

$$\mathcal{L}_\phi \supset \lambda T(\phi, \phi, \phi)^2,$$

which capture octonionic non-associativity effects in symmetry breaking dynamics.

This conceptual and mathematical integration strengthens the foundation of our model by leveraging rigorously established geometry-algebra correspondences.

6 Computability, Predictions, and Tests

- The AoO framework models the physical world as a finite computation, consistent with collider data including the LHC.
- No free fundamental scalar propagator is expected; only gauge boson decay products are observed.
- The framework predicts precise Higgs–gauge coupling constants derivable from topological strand data, enabling falsifiable experimental tests such as Higgs self-coupling measurements.
- It points toward potential extensions beyond the Standard Model through higher exceptional groups and discrete algebraic geometries.

Remark (Status of Reidemeister–Gauge Group Correspondence). In earlier formulations, the topological correspondence between the three Reidemeister moves and the Standard Model gauge groups $U(1)$, $SU(2)$, and $SU(3)$ rested primarily on qualitative topological–algebraic analogies (braid group representations, quandle cohomology, Yang–Baxter solutions) without an explicit verification of full Lie algebraic closure.

In the present AoO formulation [2, Sec. 3.3], where the complete 2×2 and 3×3 matrix–representation construction of the braid generators, commutator tables, and Jacobi identity verifications are given in detail, this correspondence is made explicit:

1. We realize the braid generators σ_1, σ_2 as concrete matrix representations:

$$\sigma_1 \mapsto e^{i\theta} \in U(1), \quad \sigma_1, \sigma_2 \mapsto SU(2) \text{ (Bureau)}, \quad \sigma_1, \sigma_2 \mapsto SU(3) \text{ (Yang–Baxter)}.$$

2. The associated Lie algebra generators are extracted and their commutators computed, verifying *Jacobi identities* and confirming closure in $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ at tree level.
3. The embedding into G_2 is exhibited explicitly, with the coset mapping $G_2/(SU(2)_L \times SU(2)_R) \rightarrow \mathcal{H}_{\text{AoO}} \subset \mathbb{O}_{\mathbb{Q}}$, a finite–dimensional quaternionic–octonionic subspace over the computable scalar base $\mathbb{Q}$. In the continuum–limit approximation this reproduces the familiar $\mathbb{C}^2$ Higgs doublet of the Standard Model, but no $\mathbb{R}$ or $\mathbb{C}$ is assumed as fundamental.

This closes the primary structural gap identified in earlier versions, which lacked an explicit verification of full Lie algebraic closure. Remaining open directions include:

- Quantum corrections via knot invariants and loop effects,
- Anomaly cancellation proofs for the full fermion content,
- Incorporation of flavor structure and potential gauge–gravity unification.

The strengthened correspondence is now both mathematically consistent at the representation–theoretic level and directly testable through the Higgs and gauge phenomenology derived from the AoO model.

7 Conclusions

This work:

1. Roots Standard Model gauge groups in fundamental knot-theoretic moves,
2. Embeds these symmetries in the exceptional Lie group G_2 via octonions,
3. Provides a novel topological–algebraic reinterpretation of the Higgs mechanism,
4. Offers a finite, computable model aligned with experimental evidence and consistent with discrete physics principles.

Future directions include exploring E_8 extensions, integrating discrete gravity into AoO, and computational implementations of quantum field theory.

Appendix: Foundational Octonionic Work

The AoO framework builds on pivotal developments in exceptional algebra and physics, notably:

- S. Catto and D. Chesley, *Twisted Octonions and Their Symmetry Groups*, *Nuclear Physics B* **331** (1989), foundational for twisted-octonion algebra under G_2 and Standard Model subgroup embeddings.
- M. Günaydin, C. Piron, and H. Ruegg, *Moufang Plane, Jordan Algebras and Octonionic Quantum Mechanics*, *Journal of Mathematical Physics* **25** (1984), pioneering connections between octonionic geometry, Jordan algebras, and quantum mechanics consistent with AoO’s discrete algebraic geometry.

References

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